

**Vehicle currencies
are made, not
given.**

What determines how
much trade one asset
intermediates?



Centre for Blockchain
Technologies

The

Making of Dominant Vehicle Currencies

Evidence from decentralised exchanges

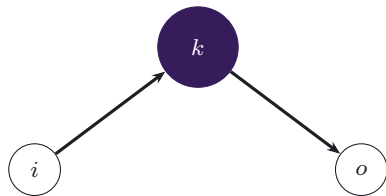
Jiahua Xu

Nanyang Blockchain Conference, NTU Singapore

Question: what makes a vehicle currency dominant?

One asset sits between many buyers and sellers. What gives it the largest share of intermediation?

1. The cheapest asset to hold and trade
2. The asset with the deepest pairing network
3. The asset embedded in routing and market design
4. Several assets, each dominant in a different margin

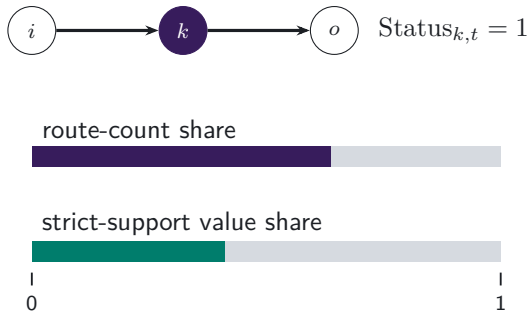


Ask the room first

Vehicle status is binary; dominance is continuous

MEASUREMENT

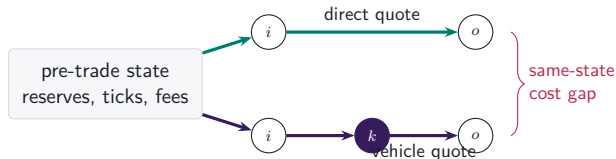
- One route through asset k gives it vehicle status.
- Dominance measures the share of intermediation that k captures.
- Count and value answer different economic questions.



Observed routes and priced alternatives share one state

IDENTIFICATION

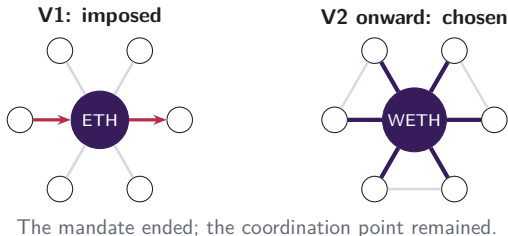
- Traditional FX records the route taken.
- On-chain pool state also prices feasible alternatives.
- The comparison holds market conditions fixed at execution.



Protocol design first imposed ETH intermediation

INSTITUTIONAL FACT

- Uniswap V1 created one exchange contract per token, paired with ETH.
- Token-to-token swaps therefore moved through ETH by construction.
- V2 allowed arbitrary pairs; the ETH-centred pairing network persisted.



The panel follows the market across six years

DATA

2,332

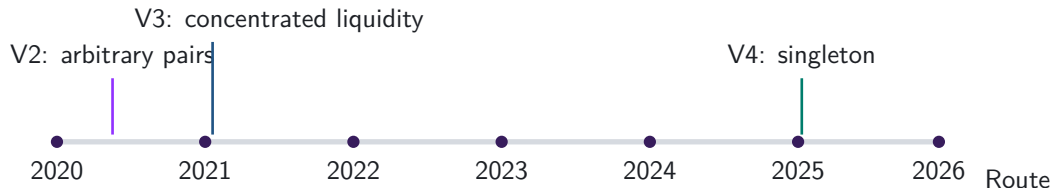
calendar days

471.6m

usable swap legs

8

DEX deployments



reconstruction is transaction-level. Daily and weekly panels are analysis clocks, not raw-data clocks.

Four measurements separate role, scale, and network position

1. Vehicle share

Fraction of route units intermediated by asset k .

2. Excess use

Intermediary share relative to endpoint demand.

3. Network position

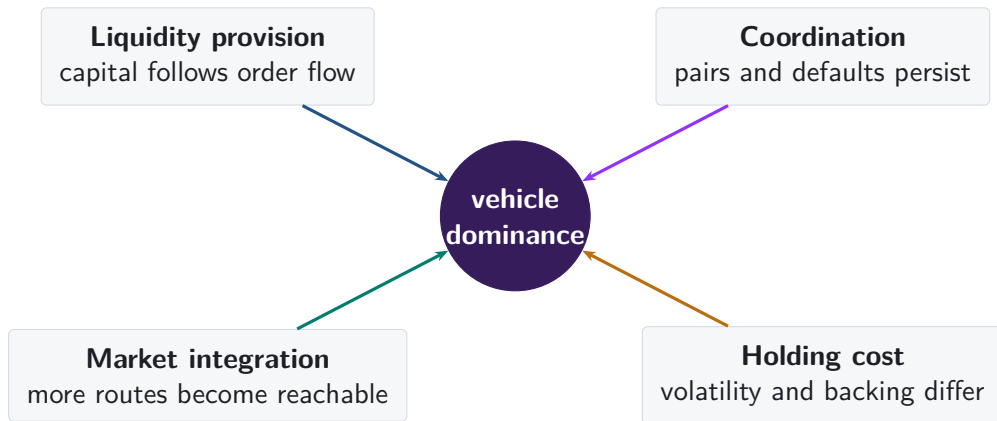
Degree, strength, and betweenness on the trading graph.

4. Execution advantage

Direct output relative to a vehicle route at one state.

A single statistic cannot stand in for all four objects.

Four forces can sustain vehicle dominance



The design measures each force separately before asking which one accounts for changes in dominance.

The empirical design separates formation, rotation, and persistence

FORMATION

Architecture changes

Imposed intermediation, arbitrary pairing, concentrated liquidity, and singleton settlement.

ROTATION

Panel variation

Count share, value share, excess use, network position, and route complexity over time.

PERSISTENCE

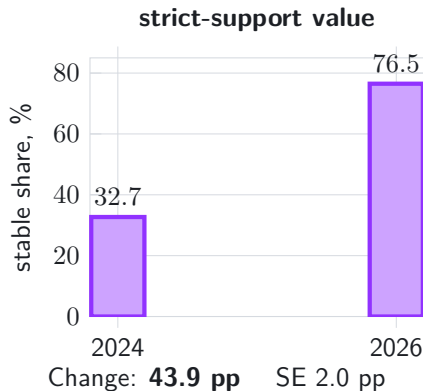
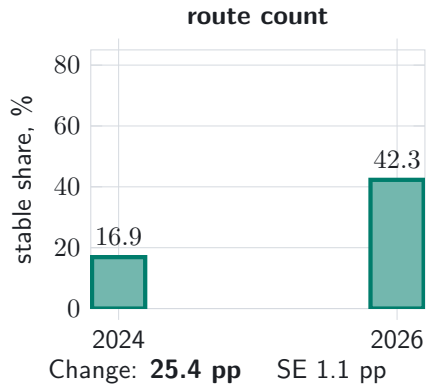
Exact alternatives

The role can be observed while another feasible route is cheaper at the same state.



Stablecoins gained vehicle share by count and by value

DESCRIPTIVE RESULT

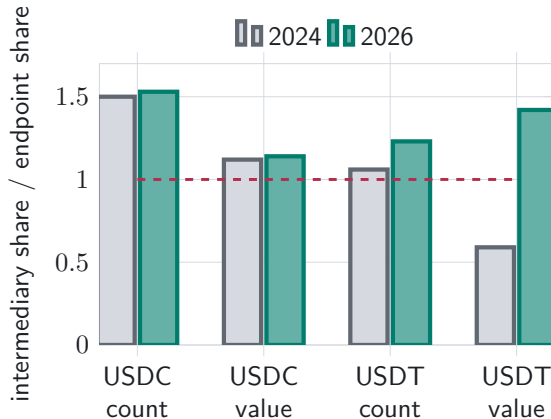


Exact two-leg routes; daily equal-weighted shares on common endpoint-year calendar support.

USDT is the transition margin

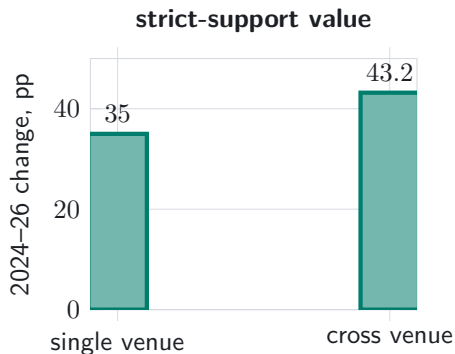
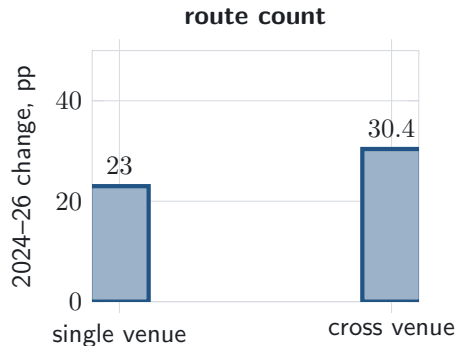
TOKEN DECOMPOSITION

- USDC already had excess intermediary use in 2024 and changed little.
- USDT moved from endpoint-heavy value to intermediary-heavy value.
- USDC and USDT account for 92.1% of the stable count-share increase.



Rotation is present within venues and stronger across venues

MARKET INTEGRATION



Cross-minus-single: **7.5 pp** by count (SE 1.3) and **8.2 pp** by value (SE 2.3).

Dominance reflects coordination, demand, and market design

1. **A protocol can create a vehicle role.**

V1 imposed ETH intermediation; the ETH-centred pairing network persisted after arbitrary pairs became possible.

2. **The dominant vehicle can rotate.**

Stablecoins gained 25.4 percentage points by route count and 43.9 points by strict-support value from 2024 to 2026.

3. **The transition is economically specific.**

USDT moved from endpoint-heavy to intermediary-heavy value, and the rotation is stronger in cross-venue routing.

Next: which force turns vehicle use into persistent dominance?

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Candidate types and backing regimes
- A3 Sample integrity and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

Robustness

- A7 Daily and weekly clocks
- A8 Count and value support
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

- A coherent $i \rightarrow k \rightarrow o$ component increments the route count once.
- Vehicle status turns on when k appears as an intermediary.
- Dominance is the share of route units or supported value carried by k .

A2. Backing regimes are dated, not permanent labels

Native platform asset

ETH, WETH; volatile settlement asset and protocol-native demand.

Fiat-reserve stablecoin

USDC, USDT; reserve and redemption design.

On-chain collateralised

DAI, LUSD, crvUSD; collateral, liquidation, and backing changes.

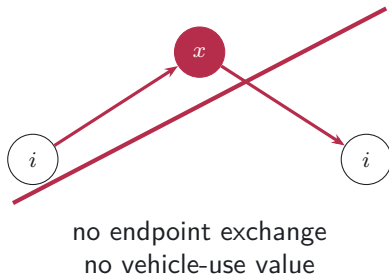
Imported store of value

WBTC, tokenised gold; wrapped claim on an external asset.

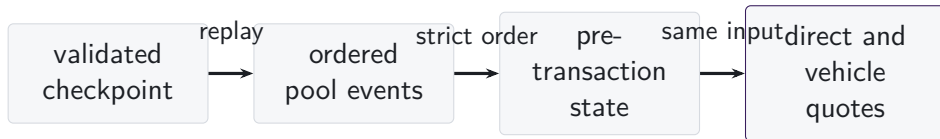
Dated backing classifications enter heterogeneity tests. Ticker categories do not substitute for backing evidence.

A3. Integrity screens protect the value denominator

- Round trips are excluded from intermediation measures.
- Token values require price consensus and explicit token identity.
- Strict-support value retains observations within the registered coherence band.



A4. State is reconstructed immediately before execution



- Checkpoints make replay parallel without changing event order within a partition.
- Quotes use the exact pool state visible before the focal transaction.
- Unsupported provider intervals remain explicit exclusions with economic-weight bounds.

A5. Each protocol family has its own state contract

Constant product

Reserves, fee, invariant. **Concentrated**

liquidity

Active liquidity, price, tick, and initialised-tick changes. **StableSwap**

families

Balances, amplification, rates, and oracle state.

Weighted pools

Vault balances, weights, scaling, and rate providers. **V4 singleton**

Pool key, exact vanilla tick state, fee, and validated hook behavior.

A venue enters an estimand only after its quantity and state contract pass validation.

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- The paired legs report equal ETH quantities in 87.4% of identified forced routes.
- The signature identifies protocol-imposed intermediation, not user choice.



217,003
forced routes

A7. Daily and weekly clocks answer different questions

- | | |
|----------------------------------|---|
| 1. Primary panel | day observations; date fixed effects |
| 2. Calendar sensitivity | day observations; week and weekday fixed effects |
| 3. Aggregation robustness | complete weeks; week fixed effects |
| 4. Transition series | exact calendar links; HAC at 1, 7, 30, and 120 days |

Weekly ratios aggregate raw numerators and denominators before division. All seven weekday anchors are reported.

A8. Count and value define separate dominance margins

Count-weighted share

$$S_{k,t}^N = \frac{N_{k,t}^I}{\sum_j N_{j,t}^I}$$

Each reconstructed route carries one unit of weight.

Value-weighted share

$$S_{k,t}^V = \frac{\text{Vol}_{k,t}^I}{\sum_j \text{Vol}_{j,t}^I}$$

Each route carries supported realised USD value.

A larger value shift can reflect route size, routing topology, or candidate composition. The decomposition tests these margins directly.

A9. USDT gains intermediary use beyond endpoint demand

Count margin

- Intermediary-minus-endpoint share rises by **2.39 pp**.
- Calendar-HAC SE: 0.89 pp.

Strict-value margin

- Gap moves from -7.13 **pp** to $+8.14$ **pp**.
- Change: 15.27 pp; calendar-HAC SE: 1.50 pp.

USDT becomes an intermediary beyond what its endpoint demand predicts.

A10. Capital, inventory, and depth are distinct quantities

Deposited capital

Funds committed to a pool; reconciled balances or audited provider TVL.

Inventory

Exact token amounts held by the pool.

Executable depth

Output near a price under the invariant and exact quote state. **Quote quality**

Realised output for notional q on executable paths.

TVL can measure capital. It is not automatically the denominator for executable depth.

A11. Integration changes both reach and route choice

Opportunity reach

- active venues
- feasible paths
- candidate vehicles

Choice within reach

- direct versus intermediated
- single- versus cross-venue
- sequential versus split routing

The interaction test asks whether vehicle rotation changes with integration while holding the opportunity cell fixed.

A12. References: vehicle currencies and market structure

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